

7.2 Trigonometric Integrals - continued

How can we use the Pythagorean trigonometric identity to relate $\sec(x)$ and $\tan(x)$? $\csc(x)$ and $\cot(x)$?

Trig Identities

- | | |
|---------------|-----------------|
| • $\tan(x) =$ | • $\tan^2(x) =$ |
| • $\sec(x) =$ | • $\sec^2(x) =$ |
| • $\cot(x) =$ | • $\cot^2(x) =$ |
| • $\csc(x) =$ | • $\csc^2(x) =$ |

How will integrals involving powers of secant and tangent be different than those involving sine and cosine?

Example 1

Evaluate $\int \tan^3(x) \sec(x) \, dx$.

Example 2

Evaluate $\int \cot^2(x) \csc^4(x) \, dx$.

Summary: Evaluating $\int \tan^m(x) \sec^n(x) dx$

Case 1: n is even (i.e. power of secant is even)

- Save one factor of $\sec^2(x)$ to be used for du .
- Convert the remaining **even powers** of secant using $\sec^2(x) = 1 + \tan^2(x)$
- You should be left with an integrand involving only tangent, multiplied by $\sec^2(x)$.
- Apply u -substitution ($u = \tan(x)$).

Case 2: m is odd (i.e. power of tangent is odd)

- Save one factor of $\sec(x) \tan(x)$ to be used for du .
- Convert the remaining **even powers** of tangent using $\tan^2(x) = \sec^2(x) - 1$
- You should be left with an integrand involving only secant, multiplied by $\sec(x) \tan(x)$.
- Apply u -substitution ($u = \sec(x)$).

NOTE: You may need two basic integrals:

- $\int \tan(u) du = \ln |\sec(u)| + C$
- $\int \sec(u) du = \ln |\sec(u) + \tan(u)| + C$

Sometimes, the cases are not clear cut. You may need to use integration by parts, or modify the integrand using another identity.

Example 3

Evaluate $\int \sec^3(x) \, dx$.

Extra example

Try this one your own before watching the video!

Evaluate $\int \tan^4(x) \, dx$.